

Qatar-2b

R-0805

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_230406 · created 2026-07-28T10:43:38+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460040.7605 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.227 +/- 0.065
Transit depth (Rp/Rs)^2	5.16 +/- 2.95 [%]
Orbital Inclination (inc)	84.73 +/- 1.67
Airmass coefficient 1 (a1)	0.42 +/- 0.2
Airmass coefficient 2 (a2)	0.46 +/- 0.3
Scatter in the residuals of the lightcurve fit is	47.73 %
Best Comparison Star	None
Optimal Aperture	4.73
Optimal Annulus	4.92
Transit Duration (day)	0.0701 +/- 0.0089

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.227 ± 0.065 vs 0.16327 (deviation 0.98σ)
Duration fit vs published	0.0701 d vs 0.0767 d (deviation 0.74σ)
Mid-time O–C	-3.6 min
Depth SNR	3.5
Residual scatter	47.73 %

Light curve

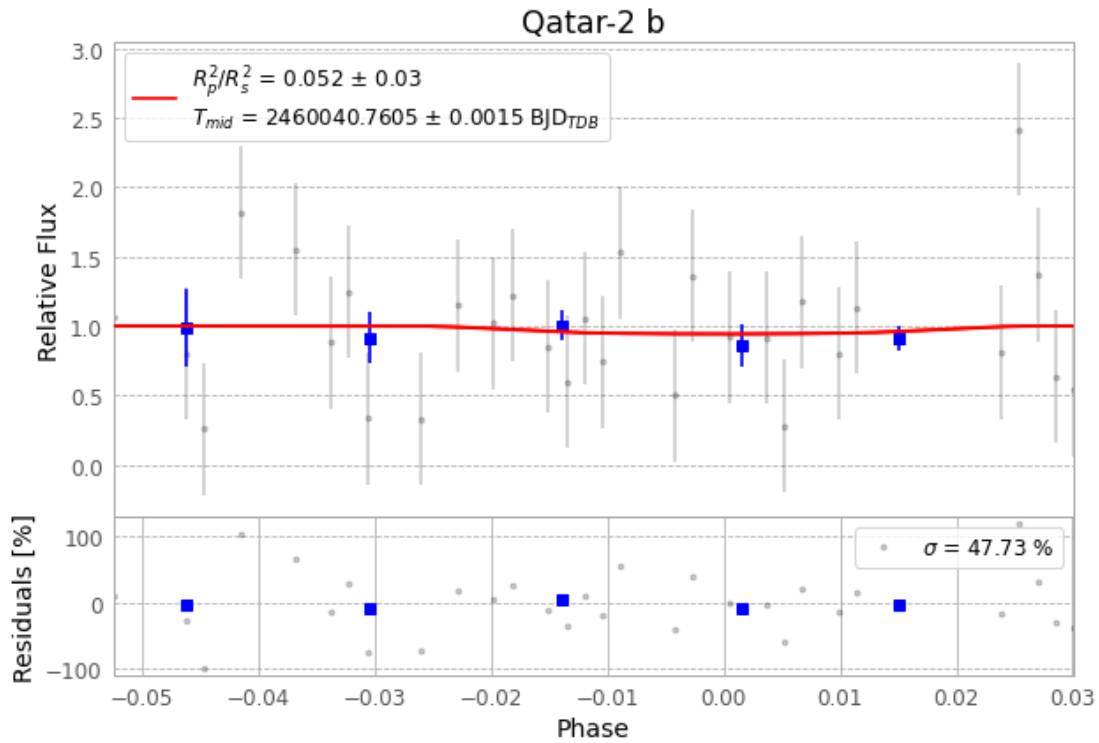


Figure 1 — FinalLightCurve_Qatar-2 b_2023-04-05.png (SHA-256 6c0fdf5c78d97adb...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [314, 381], 2 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 102182eca77af857...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

75 FITS frames (49 MB), Qatar-2230406042415.FITS → Qatar-2230406080615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-230406.log (8d8779a53ce6...), FinalLightCurve_Qatar-2 b_2023-04-05.png (6c0fdf5c78d9...), FinalLightCurve_Qatar-2 b_2023-04-05.csv (c0939445c760...)

Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 10:43Z.